

Complete solutions to all problems are required. Models should be specified, approximations and conclusions should be specified and motivated.

Problems 1, 2 and 3 can give a maximum of 10 points each. Problems 4, 5 and 6 can give a maximum of 20 points each. The maximum total on the exam is 90 points. The limit for Pass (P) is at least 45 points, and the limit for High Pass (HP) is at least 67 points.

Only the paper sheets provided by the institute in the exam room shall be used, both for solutions and for sketches/notes.

Each problem solution shall start at the top of a new paper.

Write your ID number on each paper that you hand in.

Red pencil or pen is not allowed.

Only statistical tables provided by the institute are allowed. Electronic calculators are allowed. No other tables or formula sheets are allowed

1. Given following 5 independent observations from a continuous random variable  $X$ , taking values in  $\mathbf{R}$

0.509, 0.971, 0.0493, 0.623, 0.0488.

The random variable  $X$  has expectation  $E(X) = m(\theta) = 1 - \theta^2$ . The parameter space is  $\Theta = (-1, 1)$ . Find the least squares estimator of  $\theta$ . [10p]

2. In two separate and statistically independent experiments the researchers want to check if two individuals A and B have (as they claim) the ability to foresee events in the future. In the first experiment one asks A to foresee the outcome of a fair coin toss, and out of 10 tries A succeeds to foresee the outcome correctly 4 times. In the second experiment one asks B to foresee the outcome of a toss of (another) fair coin and out of 21 coin tosses B succeeds to foresee the outcome correctly 15 times.

- (a) Assuming that A and B are equally good at foreseeing the future, find the maximum likelihood estimator of the their probability to foresee the future. [5p]
- (b) Assuming that A and B are equally good at foreseeing the future, perform an exact test of

$H_0$  : A and B do not have an ability to correctly foresee the future,

$H_1$  : A and B do have an ability to correctly foresee the future.

[5p]

3. Let  $X$  be a random variable with density function

$$f(x) = \begin{cases} \frac{1}{4} \binom{25}{-x} \theta^{-x} (1 - \theta)^{25+x}, & x = -25, \dots, -2, -1, 0, \\ \frac{3}{4} \theta e^{-\theta(x-1)}, & x \geq 1, \end{cases}$$

with respect to the measure  $\mu_1(x)1\{x \leq 0\} + \mu_2(x)1\{x > 0\}$ , with  $\mu_1$  counting measure and  $\mu_2$  length measure. The parameter space is  $\Theta = [0, 1]$ . We have the following i.i.d. observations of  $X$

-11, -15, -12, -11, 2.42, 3.03, 2.65, 1.88, 1.03.

- (a) Find the maximum likelihood estimator of  $\theta$ . [5p]  
 (b) Let  $\eta = h(\theta) = P_\theta(-3 \leq X \leq 3 | X \geq -1)$ . Find the plug-in estimator of  $\eta$ . [5p]
4. Suppose that  $X_1$  is  $Po(1/\theta)$  distributed and  $X_2$  is  $Po(2/\theta)$  distributed and suppose that we have the independent observations of  $X_1$  and  $X_2$

$$x_1 = 7, x_2 = 23.$$

The parameter space is  $[0, \infty)$ .

- (a) Find the least squares estimator of  $\theta$ . [5p]  
 (b) Based on the data, perform an exact test of the hypotheses

$$H_0 : \theta \geq 1/5,$$

$$H_1 : \theta < 1/5.$$

on significance level 0.01. [5p]

- (c) You are now given the following test:  $(T, C)$  where  $T(x_1, x_2) = x_1 + x_2$ ,  $C = [35, \infty)$ . What is the approximate power of this test, using an adequate approximation, at  $\theta = 1/15$ ? [10p]

5. Suppose that one makes independent pH measurements of water acidity in two rivers A and B. The measurements may be assumed to follow the Normal (Gaussian) distribution, with possibly different expectations and variances. All measurements are assumed independent. The following are the measurements data.

dataset A     3.55, 0.184, 2.28, -0.714,

dataset B     -1.97, 0.313, 0.334, 1.784, 0.669.

- (a) Let  $\sigma_A^2$  and  $\sigma_B^2$  denote that variances in the two data sets, and test the hypotheses

$$H_0 : \sigma_A^2 \leq \sigma_B^2,$$

$$H_1 : \sigma_A^2 > \sigma_B^2,$$

on significance level 0.05. [10p]

- (b) Assuming that  $\sigma_A^2 = \sigma_B^2 =: \sigma^2$ , construct a lower bounded confidence interval for  $e^{-\sigma^2}$  with confidence grade 0.99. [10p]

6. Suppose that  $T$  is a continuous random variable and let  $S(t) = 1 - F(t)$  be it's survival function. Independent observations of  $T$  are

3.81, 2.64, -2.38, -3.69, 1.45, -4.10, 1.13, -0.679, -0.0603,

-0.502, -4.79, 1.48, -0.480, 2.45, 3.10, -4.80, -2.51, 1.40, 2.67, 4.25.

- (a) Perform an exact test of the hypotheses

$$H_0 : S(3) \leq 1/10,$$

$$H_1 : S(3) > 1/10.$$

on significance level 0.05. [10p]

- (b) Using an appropriate approximation, construct an approximate test of the above hypotheses and calculate the constructed test's power at  $S(3)=1/5$ . [10p]